



GE Medical Systems

Service Note

**Direction 2333261-100
Revision 0**

**GE Medical Systems
Recon (ICE) Box Troubleshooting, T-1446**

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Revision History

Revision	Date	Reason for change
0	021202	Initial Release

Service Note, T-1446

Effectivity: CT systems Using the following Operator consoles:

- 2180551 LS (QX/i) Octane I w/ O2 and H1.1 SW
- 2180551-2 LS (QX/i) Octane I w/o O2
- 2266832 LS Plus with Octane I
- 2266832-2 LS Plus with Octane II
- 2304732-3 LS Ultra with Octane I
- 2304732-4 LS Ultra with Octane II

Problem: Motorola and DAS Interface Processor Bd. Usage Increasing

RIP and DIP boards account for the highest number of FRUs changed out in-warranty and/or received as a DOA part. Two issues have been identified as root causes; mis-identification and incorrect fault isolation.

Solution: Console Compatibility Matrix & T/S Overview Required

In addition to providing the following matrix and T/S information, System manuals for Lightspeed system have update to include the following information.

Section 1 - RIP Board Compatibility

In the material that follows, a matrix for matching both RIP and DIP boards to the correct console type is shown. Use Table 1 below to identify which model of RIP you should be using.

GE Part #	Motorola Part #	Used On Console Part #	(J15) Jumper Settings	(J16) Jumper Setting	Memory Size	Description
2207232* 2197234* 2197234-3	MVME2306	2180551	2-3	2-3	32MB	LS (QX/i) Octane I w/ O2 and H1.1 SW
2197234-4	MVME2306	2180551	2-3	2-3	32MB	LS (QX/i) Octane I w/ O2 and H1.2 SW
2197234-2	MVME2308	2180551-2	2-3	2-3	128MB	LS (QX/i) Octane I w/o O2
2197234-2	MVME2300	2266832 2266832-2	2-3	2-3	128MB	LS Plus with Octane I
2197234-2	MVME2300	2304732 2304732-2	2-3	2-3	128MB	LS Plus with Octane II
2197234-2	MVME2300	2266832-3 2266832-4	2-3	2-3	128MB	LS Ultra with Octane I
2197234-2	MVME2300	2304732-3 2304732-4	2-3	2-3	128MB	LS Ultra with Octane II

* Obsolete product

Table 1 RIP Board Compatibility Matrix

Section 2 - DIP Board Compatibility

DIP Board

Table 2, below, can be used to help identify which model DIP board is used on which LightSpeed family console.

GE Part #	Used On Console Part #	Description
2160800	2180551	QX/i with Octane I w/ O2 and H1.1 (DIP Board found in O2 computer)
2160800	2180551	QX/i with Octane I w/ O2 and H1.2 (DIP Board found in O2 computer)
2216666	2180551-2	QX/i with Octane I w/o O2
2245261	2266832 2266832-2	Plus with Octane I
2245261	2304732 2304732-2	Plus with Octane II
2259530	2266832-3 2266832-4	Ultra with Octane I
2259530	2304732-3 2304732-4	Ultra with Octane II

Table 2 DIP Board Compatibility Matrix

Section 3 - Confidence Tests

A finite number of tests exist to verify the operation and reliability of the Recon (ICE) box. Every board and I/O path in the Recon box has an associated test that exercises it. Figure 1 illustrates the relationship between the steps that follow and their coverage. They represent the minimal number of test that should be run to ensure confidence in the box.

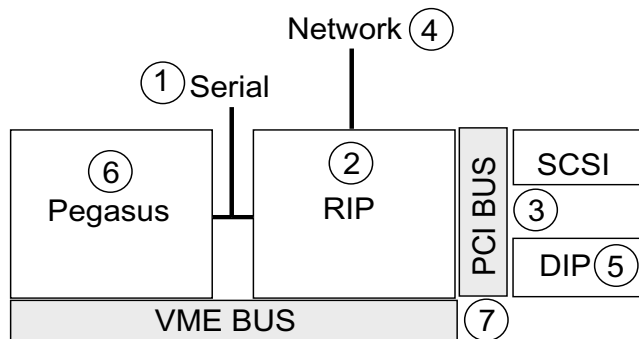


Figure 1 - Recon (ICE BOX) Diagnostic Coverage

If you have difficulty performing any of the steps below, please see the appropriate system service manual for details on the execution of commands. *"They can vary from system to system."*

- 1.) Verify Serial Communications path is operating properly from computer to recon box by using the cu command.
 - A.) **cu PIG** and verify you see VxWorks ("~." to return to OC)
 - B.) **cu ICE**
- 2.) Log-n to the ICE and run a full set of Motorola Diags (**ice fulldiags**)

- 3.) Check and verify that the SCSI card and DIP board are recognized by the RIP board
 - A.) Execute **pciDeviceShow** command on the ICE; device 10 is listed.
 - B.) Execute **scsiShow** command on the ICE; drive information is displayed
- 4.) Check communications between the host and Recon box.
 - A.) Ping the Host computer from the ICE board; **ping <hostname>**)
 - B.) Ping the ICE from the host computer; **ping <hostname>**)
 - C.) From the ICE box, check that the raw_data directory and subdirectories exists;
ls "/raw_data"
- 5.) Run the DIP diags, including DMA test
Internal loopback on DIP boards using Fiber is not implemented. You must use a loopback cable.
- 6.) Run the Pegasus Image Generator (PIG) diags.
- 7.) Run recon data path test.



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